Formula 1 Podium Prediction

Milestone 1 analytical report · Kaggle *Formula 1 World Championship (1950–2024)* · companion to `F1_Podium_Prediction.ipynb`

> **Task.** For every driver in every race, predict whether they finish on the podium (top 3). The model uses only information known **after qualifying and before the start**. The modeling table has 25,305 driver-race starts from 1950–2024, with a podium rate of 13.4 %.

1. Data and cleaning

The 14 linked Ergast CSV tables were loaded as raw text. `\N` sentinels were converted to missing values and columns were typed: numbers, dates, and lap times converted to seconds. Inconsistent labels were unified (for example *United States* → *USA*). All primary keys are unique and complete, and every foreign key matches, with 0 orphans. 176 duplicate (race, driver) rows come from 1950s–60s shared drives; for each, the row with the driver's best finishing position was kept. Entries that never started (DNQ, DNPQ, 107 % rule) were removed.

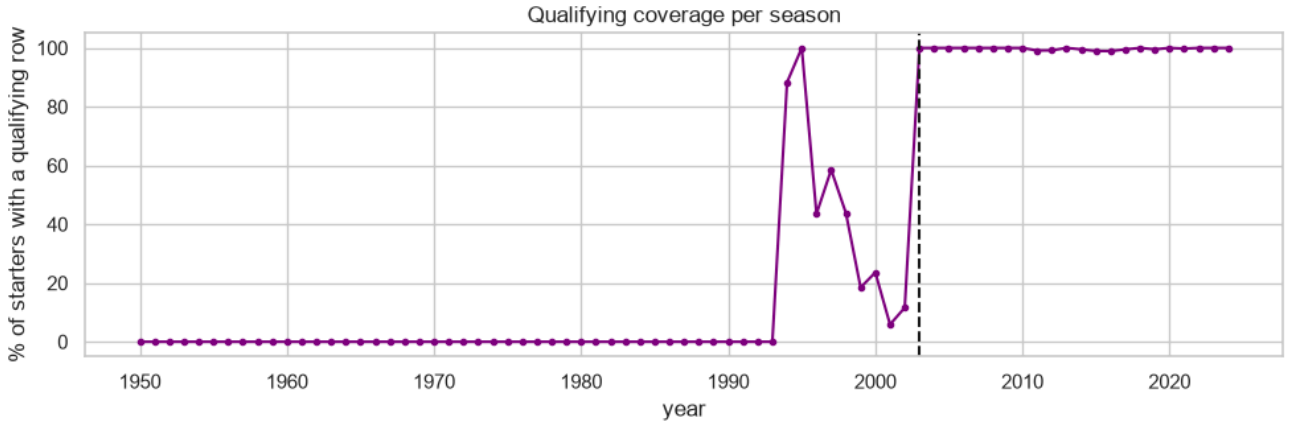


Figure 1. Qualifying coverage by season: missing before 1994, patchy until 2002, complete from 2003.

2. Data-engineering questions

Q1 — Which circuits convert a front-row start into a podium most often, and does this change after 2014?

Front-row starters (grid 1–2) reached the podium **68 %** of the time in 2003–2013 and **76 %** in 2014–2024, reflecting the dominant hybrid-era teams. The circuit rankings in the two eras agree only weakly (Spearman $\rho \approx 0.30$, $p \approx 0.29$, 14 circuits), so **the ranking changes after 2014**. Conversion depends more on each era's competitive order than on the track. Measured by qualifying position instead of grid, the hybrid-era rate is 73.7 %. The two definitions differ because of grid penalties; for example, Interlagos converts 79 % by grid but 65 % by qualifying.

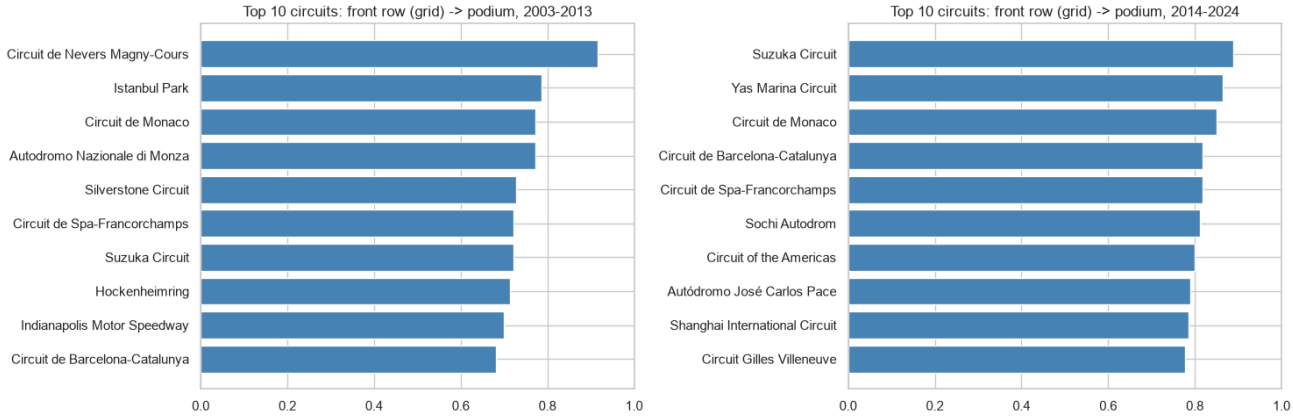


Figure 2. Top circuits by front-row podium conversion, 2003–2013 vs 2014–2024.

Q2 — Do drivers podium more often at home, after controlling for starting position?

Without controls, home drivers podium *less* (11.5 % vs 13.8 %). Within each grid bucket, the home and away rates are similar. A logistic regression `podium ~ home + grid bucket + decade` gives an odds ratio of **0.92 (95 % CI 0.78–1.08, p ≈ 0.29)**. The interval contains 1, so there is **no evidence of a home advantage**. This is an association, and nationality is only a rough proxy for "home".

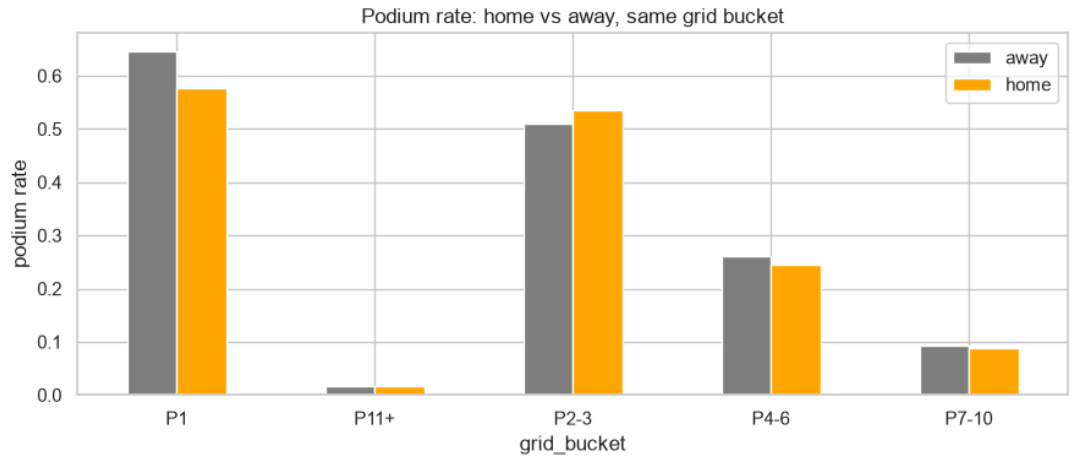


Figure 3. Podium rate at home vs away within grid buckets.

Q3 — Which constructors have the highest mechanical-retirement rate, 2014–2021 vs 2022–2024?

Mechanical retirements are retirements caused by failures such as engine, gearbox or hydraulics; crashes and disqualifications are excluded. The rate is these retirements divided by race starts, for teams with at least 20 starts. The average fell from **9.5 %** to **6.0 %**. In 2014–2021 the highest rates were Caterham (23.5 %), Lotus F1 (17.1 %) and Toro Rosso (16.1 %). In 2022–2024 they were Alfa Romeo (12.5 %), Alpine (11.0 %) and AlphaTauri (8.0 %). Teams have only about 60–140 starts each in 2022–2024, so small differences are not significant. `status` is post-race information, so this analysis is descriptive only and is never used as a feature.

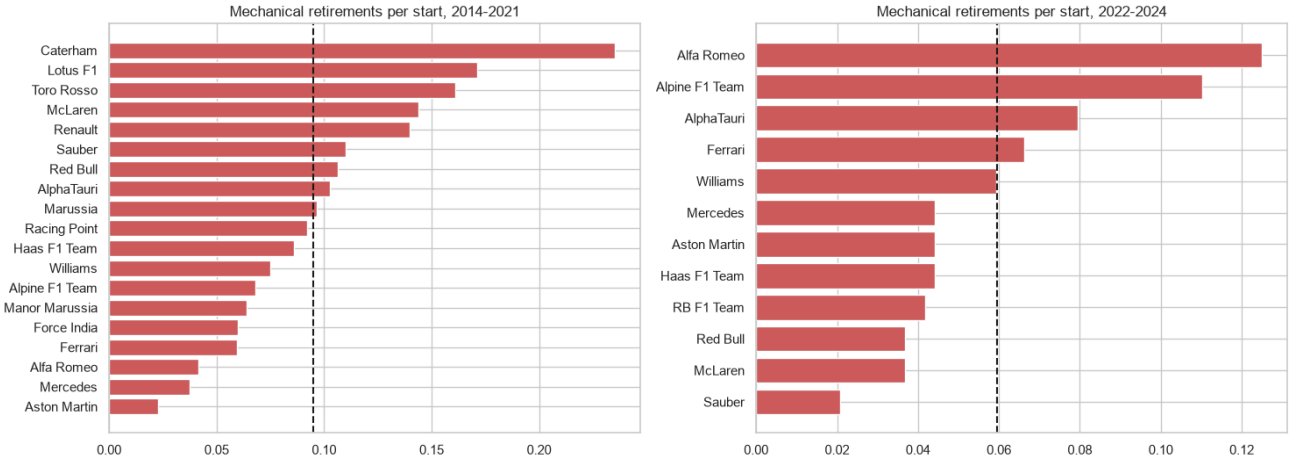


Figure 4. Mechanical-retirement rate per constructor in the two periods.

3. Features and the pre-race cutoff

Every feature must be known before the start. The current race's result, points, laps, status, lap times and pit stops are excluded, as are standings that include the current race. To show why, `points` from the same race reaches an AUC of 0.975: almost perfect, because it encodes the answer.

Group	Features	Why it is valid before the start
grid	`grid_feat`, `pitlane_start`	The starting slot is fixed after qualifying and penalties.
qualifying	`quali_pos`, `quali_gap_pct`, `has_quali`	Qualifying ends before the race. The gap to pole in % removes circuit-length effects, and `has_quali` flags rows where qualifying data does not exist.
driver form	`drv_podiums_last5`, `drv_avg_finish_last5`	Rolling over the previous 5 races only (`shift(1)` before rolling); verified on 100 random rows, with 0 mismatches.
team form	`team_podiums_last5`	The same past-only rolling window, applied to the constructor.
standings	`drv_points_share`, `drv_champ_pos`, `team_champ_pos`	Taken from standings after the *previous* round of the same season.
context	`driver_age`, `home_race`	Known from date of birth, nationality and circuit country.

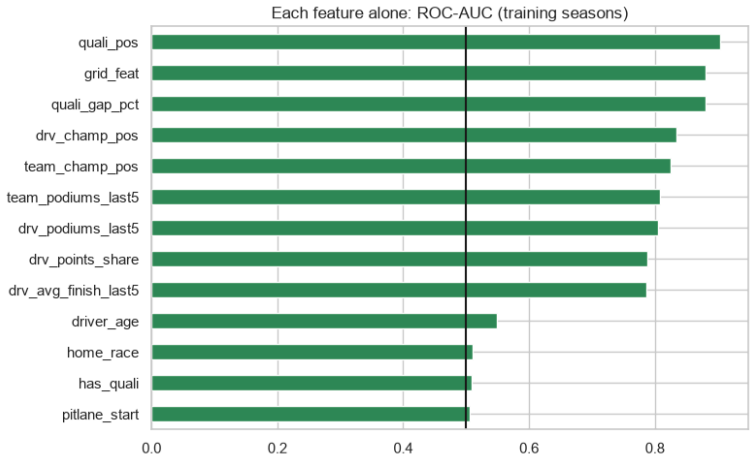


Figure 5. Single-feature ROC-AUC on the training seasons. Grid alone reaches 0.881.

4. Split and pre-processing

The data is split by season so that the model is always tested on the future: **train 1950–2017** (22,326 rows), **validation 2018–2021** (1,620) and **test 2022–2024** (1,359; new technical rules from 2022). Missing values are filled with the training median, and features are standardised for logistic regression and the neural network. Both steps are fit on train only.

Experiment (logistic regression)	val PR-AUC	test PR-AUC
E1: fit on train only (correct)	0.7059	0.7102
E1: fit on all data before split (leaky)	0.7057	0.7095
E2: impute → scale	0.7059	0.7102
E2: scale → impute	0.7059	0.7096

The differences appear only in the 4th decimal. The leaky version is still wrong in principle, because it uses test statistics during training. A third experiment trained only on 2003–2017, where qualifying data is complete. It changed PR-AUC by less than 0.01 for both models (LR 0.706 → 0.700 on validation; RF 0.725 → 0.729), so all seasons are kept and `has_quali` flags the old rows.

5. Models and metric choice

Four models are trained on the same rows and features: **logistic regression** (baseline), a **random forest** (12 configurations searched), **gradient boosting** (3 setups) and a **shallow neural network** (Keras, 3 architectures × 2 L2 strengths, early stopping). Hyperparameters and decision thresholds are chosen on validation only. Each threshold maximises validation F1, and the test set is only reported.

Metric. Only about one row in seven is a podium, so accuracy is misleading: always predicting "no podium" scores 86 %. ROC-AUC is also inflated, because most of its credit comes from ranking the many non-podium rows correctly. The primary metric is **PR-AUC** (average precision), which measures how well real podium finishers are ranked at the top. F1 at the chosen threshold is the secondary metric, because it reflects the yes/no decision a user would act on.

Model	Val PR-AUC	Test PR-AUC	Val F1	Test F1	Test ROC-AUC	Threshold
Logistic regression (baseline)	0.7059	**0.7102**	**0.7229**	0.6816	0.9337	0.346
Random forest	0.7254	0.7079	0.7195	0.6667	0.9408	0.449
Gradient boosting	0.7094	0.7093	0.7140	**0.6847**	**0.9422**	0.418
Shallow neural network	**0.7264**	0.6955	0.7203	0.6707	0.9381	0.289

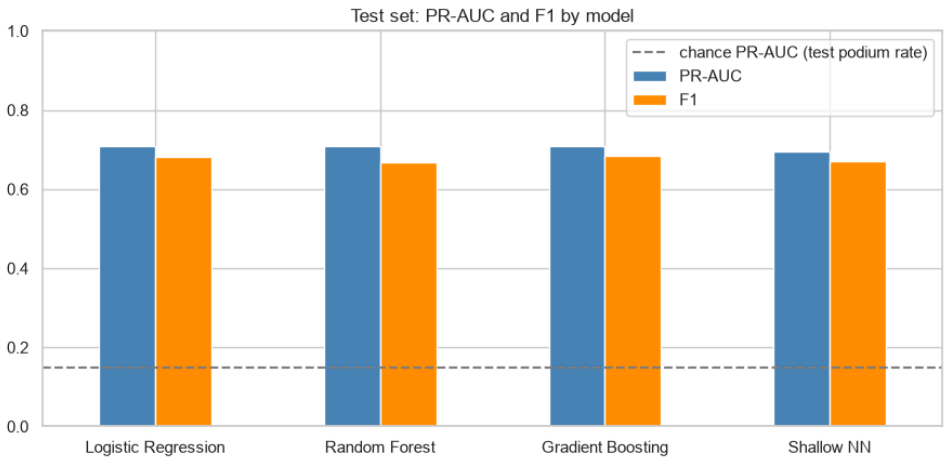


Figure 6. Test PR-AUC and F1 per model, with the podium rate shown as the chance line.

Reading. All four models fall within about 0.02 PR-AUC of each other. The neural network is best on validation (0.7264), just ahead of the random forest (0.7254). On test, however, the logistic-regression baseline is marginally best (0.7102). The non-linear models therefore do not clearly beat the baseline: most of the signal is the near-monotonic grid effect, which a linear model already captures. The random forest is used for explanation and inference because it is 0.001 behind the validation leader and supports exact tree-based SHAP. This does not mean it is statistically better.

6. Feature-group ablation

Each group was removed in turn and the model retrained with the same hyperparameters and a re-tuned threshold. The random forest was averaged over 3 seeds; differences below about 0.005 are noise. The full random forest scores 0.725 PR-AUC on validation and 0.709 on test.

Removed group	RF val drop	RF test drop	LR val drop	LR test drop	RF test PR-AUC, this group only
grid	**0.020**	**0.034**	**0.024**	**0.024**	0.647
qualifying	0.003	−0.007	0.006	0.007	0.646
driver form	0.004	−0.005	0.000	−0.004	0.550
team form	0.000	0.000	−0.001	0.006	0.480
standings	0.004	0.008	−0.002	0.006	0.562
context	0.005	0.004	0.002	0.001	0.144

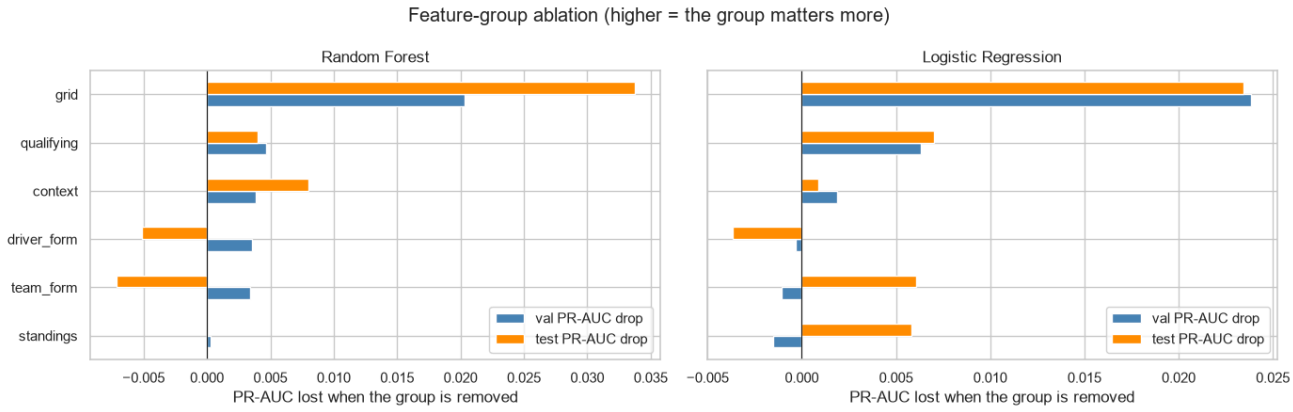


Figure 7. PR-AUC drop when each group is removed (validation vs test, RF and LR).

Grid is the only group whose removal clearly hurts in both models and both splits. Qualifying, form and standings are each strong on their own but **redundant** with grid and with each other, so removing any one of them costs almost nothing. Context is no better than chance on its own (0.144, about the podium rate), which agrees with Q2. Small negative test drops come from a single, small test split and are not taken as a reason to remove features.

7. Explainability

Permutation importance (validation, 20 shuffles each): shuffling `grid_feat` lowers PR-AUC by 0.2115. Next come qualifying gap (0.013), points share (0.011), recent podium rate (0.010) and championship position (0.008). Most non-grid drops are smaller than their shuffle variability. **SHAP** (TreeExplainer, 300 validation rows) ranks grid first as well (mean |SHAP| 0.090), followed by recent podium rate (0.026), championship position (0.022) and points share (0.019). A low grid number, a high recent podium rate, a high points share and a good championship position all push predictions towards a podium. The two methods measure different things (loss of score vs contribution to each prediction), so their middle rankings differ. Correlated features also share credit.

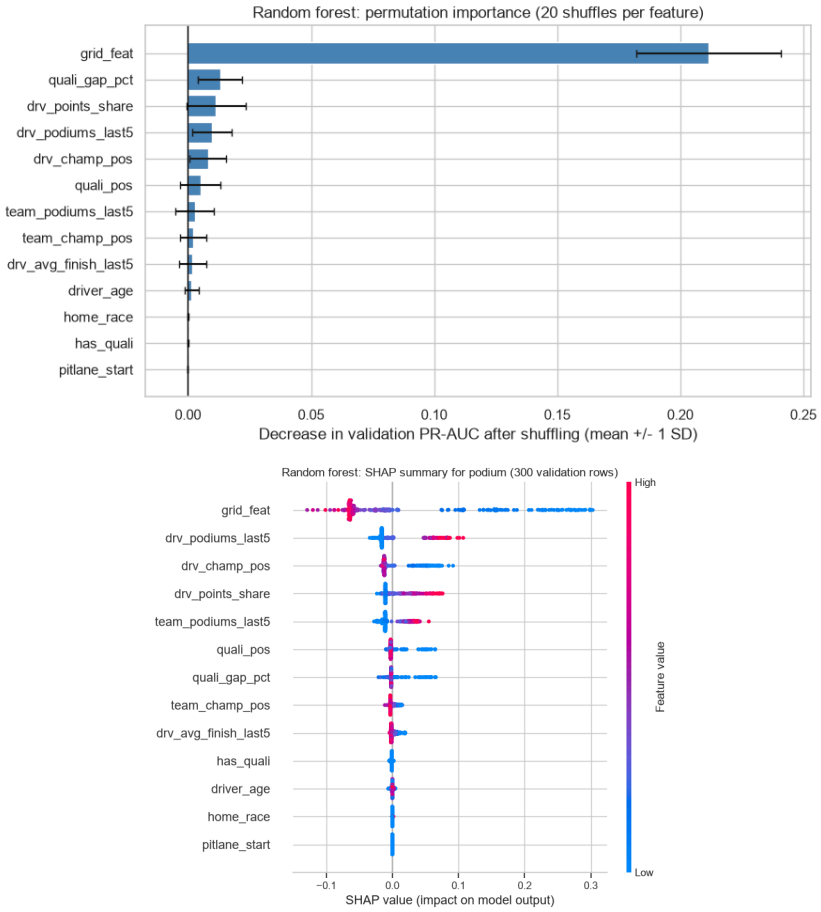


Figure 8. Permutation importance (left) and SHAP summary (right) for the random forest on validation data.

Local explanation (LIME). The case explained is Valtteri Bottas at the 2019 Mexican Grand Prix. It was chosen because its forest probability (0.449) is the closest to the threshold, without looking at the outcome. The strongest supporting conditions are a top-six start (+0.231) and a recent podium rate above 0.20 (+0.057), followed by points share and championship position. LIME's local surrogate predicts 0.439, close to the forest's value, with a local R^2 of 0.86. Bottas did finish on the podium.

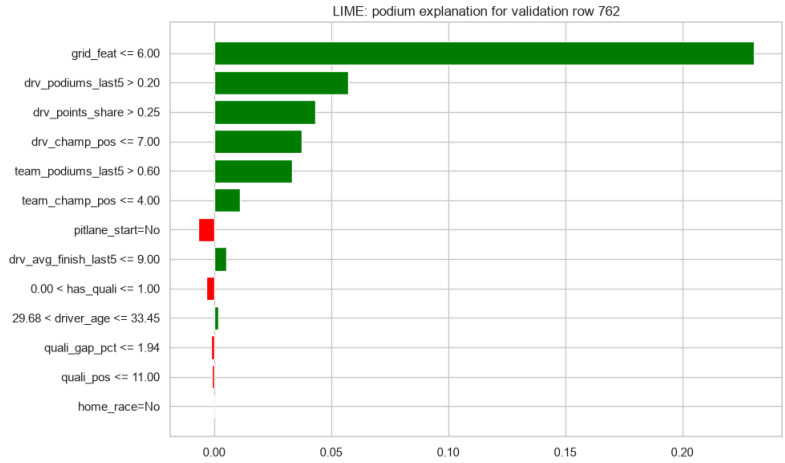


Figure 9. LIME explanation of the borderline Bottas, Mexico 2019 prediction.

> **Causal caveat.** Importance, SHAP and LIME describe what the *fitted model* relies on, not what causes podiums. Grid position, for example, partly stands in for car speed. Features are correlated, permutation and LIME create unrealistic combinations, and none of these methods supports a claim like "starting one place higher causes X % more podiums".

8. Inference

`predict_podium(record)` takes a single driver-race record (a dict, a Series or a one-row DataFrame). It fills missing or absent features with the training-fitted median imputer, applies the random forest and its validation threshold (0.449), and returns the probability, the decision and the list of imputed features. The notebook demonstrates it on a complete test record and on the same record with three values blanked and one left out. Median imputation pulls an incomplete record towards an average driver, so predictions with many imputed features should be trusted less.

9. Limitations

- **Qualifying-coverage drift:** qualifying data is missing for about 59 % of rows, mostly before 2003. The model sees imputed values plus a `has_quali` flag for old seasons, and real values for modern ones.
- **Era sensitivity:** one model covers 1950–2017, across different points systems, reliability (the DNF rate fell from about 50 % to 15 %), field sizes and regulations. The 2022 rule change separates training from test.
- **Small evaluation sets:** there is one time-based split, and validation and test have only about 1,600 and 1,400 rows. Differences of about 0.01 PR-AUC between models are within noise.
- **Correlated features:** grid and qualifying position correlate at about 0.96, which makes coefficients and importances unstable.
- **Independent predictions:** the models do not enforce exactly three podium finishers per race.
- **No causal claims:** the explanations describe the model only (see Section 7).